

REPRESENTATION THEORY OF AFFINE LIE ALGEBRAS

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ABSTRACT. In this report, we study the basics of the representation theory of affine Lie algebras. In particular, we work through some basic examples in the case of $\mathfrak{g} = \mathfrak{sl}_2$, and compare with the representation theory of classical Lie algebras.

1. INTRODUCTION

The structure theory and representation theory of simple finite Lie algebras is widely considered as complete with numerous applications in physics and other areas of mathematics. However, the strict rules defining such Lie algebras did not allow much freedom, but their generalisations to infinite-dimensions led to deeper theories. For example, affine Lie algebras possess additional structure leading to Wess-Zumino-Novikov-Witten (WZNW) conformal field theory discovered by E. Witten. In this report, we will only study the foundations of the representation theory of affine Lie algebras, which closely mirrors the representation theory of its finite-dimensional counterpart.

In Section 2, we recall the construction of affine Lie algebras. We then show that they admit a root space decomposition in Section 3 mirroring classical Lie theory. Lastly, Section 4 will discuss highest weight representations of affine Lie algebras, in particular the irreducibility of Verma and Weyl modules. Although these statements require too much background to prove in the scope of this report, our main goal will be to take $\mathfrak{sl}_2$ as a running example and play with some simple examples of Verma and Weyl modules.

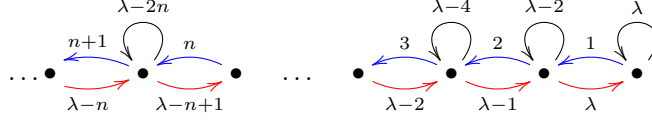
The exposition of this report follows [EFK98, Lecture 2] closely. The examples for Verma and Weyl modules inspired by [Kle], which have been expanded upon. Familiarity with the representation theory of classical Lie algebras is required for this report. See [Hum21, Chapter 0] for a reminder of the representation theory of classical Lie algebras, or see below for the basics of the representation theory of $\mathfrak{sl}_2$ needed for this report.

PRELIMINARIES

Notation. Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra over $\mathbb{C}$ and $\mathfrak{h} \subset \mathfrak{g}$ a Cartan subalgebra. Fix a set $\Delta = \{\alpha_1, \dots, \alpha_n\} \subset \mathfrak{h}^*$ of simple roots contained in a set Φ^+ of positive roots. Let $\Phi^- := -\Phi^+$ be the set of negative roots. This then fixes a triangular decomposition

$$\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{h} \oplus \mathfrak{n}^+.$$

Analogously, define the coroots $\Phi^\vee$ with their corresponding subsets $(\Phi^\vee)^+$ and $\Delta^\vee$. Let $\alpha_i^\vee$ be the simple coroot defined by $\langle \alpha_i, \alpha_i \rangle = 2$. Here, $\langle -, - \rangle$ is a non-degenerate bilinear form on $\mathfrak{h}^*$ defined by $\langle \lambda, \alpha, \rangle := \lambda(\alpha^\vee)$. A weight $\lambda \in \mathfrak{h}^*$ is

FIGURE 1. Verma module for $\mathfrak{sl}_2$

integral if $\langle \lambda, \alpha \rangle \in \mathbb{Z}$ for all $\alpha \in \Phi$ and dominant integral if $\langle \lambda, \alpha \rangle \in \mathbb{Z}_{>0}$. We define a partial ordering on $\mathfrak{h}^*$, where $\mu \leq \lambda$ if $\lambda - \mu \in \Phi^+$. Lastly, let $U(\mathfrak{g})$ be the universal enveloping algebra of $\mathfrak{g}$.

Primer on $\mathfrak{sl}_2$ basics. Let $\mathfrak{g} = \mathfrak{sl}_2$, the Lie algebra of 2×2 traceless matrices with complex entries. It has a basis

$$e := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad h := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad f := \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Lie algebra's can act on themselves via the adjoint action, so the Lie algebra structure of $\mathfrak{sl}_2$ is generated by the commutators for its basis elements given by

$$[h, e] = 2e, [h, f] = -2f, \text{ and } [e, f] = h.$$

The set of roots of $\mathfrak{sl}_2$ is $\Phi = \{\pm\alpha\}$, where $\alpha: \mathfrak{h} \rightarrow \mathbb{C}$ is given by $\alpha(h) = 2$, and the simple roots are $\Delta = \{\alpha\}$. The only positive root is α and the only negative root is $-\alpha$. We denote by $\rho = \frac{1}{2}\alpha$ the half-sum of positive roots.

The $\mathfrak{sl}_2$ -module defined by

$$\Delta(\lambda) := \text{Ind}_{\mathfrak{h} \oplus \mathfrak{n}^+}^{\mathfrak{g}} U(\mathfrak{g}) \otimes_{U(\mathfrak{h} \oplus \mathfrak{n}^+)} \mathbb{C}_\lambda$$

is the Verma module corresponding to λ . Here $\mathbb{C}_\lambda$ is the 1-dimensional $\mathfrak{h}$ -module with the action $h \cdot v = \lambda(h)v$ for all $h \in \mathfrak{h}$. By the Poincaré-Birkhoff-Witt theorem, $U(\mathfrak{g}) \cong U(\mathfrak{n}^-) \otimes U(\mathfrak{b})$, allowing us to write $\Delta(\lambda) \cong U(\mathfrak{n}^-) \otimes \mathbb{C}_\lambda$ as a left $U(\mathfrak{n}^-)$ module. Hence $\Delta(\lambda)$ is a free $U(\mathfrak{n}^-)$ -module of rank one with basis $\{\frac{1}{i!} f^i \otimes v \mid i \in \mathbb{Z}_{\geq 0}\}$ for some basis element $v \in \mathbb{C}_\lambda$. Pictorially,

where the black arrows represents the weights under the action of h , the blue arrows depicts multiplication by the scalar on top under the action of f , and the red arrows by the action of e . It follows that if $\lambda \in \mathbb{Z}_{<0}$, then $\Delta(\lambda)$ is simple since otherwise, if λ is integral, there will be n such that $\lambda - n = 0$.

2. AFFINE LIE ALGEBRAS

Let us recall the construction of the affine Lie algebra $\tilde{\mathfrak{g}}$.

Definition 2.1 (Loop algebra). The *loop algebra* $\mathcal{L}(\mathfrak{g})$ of $\mathfrak{g}$ is the Lie algebra with underlying vector space

$$\mathcal{L}(\mathfrak{g}) := \mathfrak{g} \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}],$$

with Lie bracket given by

$$[x \otimes f, y \otimes g] := [x, y] \otimes fg$$

for all $x, y \in \mathfrak{g}$ and $f, g \in \mathbb{C}[t, t^{-1}]$.

The loop algebra can also be understood topologically as the trigonometric polynomial maps $S^1 \rightarrow \mathfrak{g}$ spanned by $\{xe^{in\theta} \mid x \in \mathfrak{g}, n \geq 0\}$, with the Lie bracket evaluated pointwise. These two descriptions are isomorphic.

Recall that the central extension of $\mathfrak{g}$ is a Lie algebra $\mathfrak{e}$ such that the sequence

$$0 \longrightarrow \mathfrak{a} \longrightarrow \mathfrak{e} \longrightarrow \mathfrak{g} \longrightarrow 0$$

is exact, where $\mathfrak{a}$ is in the center of $\mathfrak{e}$. If we fix an invariant symmetric bilinear form $\langle -, - \rangle$ on $\mathfrak{g}$, then there exists a central extension of $\mathcal{L}(\mathfrak{g})$ given by the vector space $\hat{\mathfrak{g}} := \mathcal{L}(\mathfrak{g}) \oplus \mathbb{C}c$, endowed with Lie bracket

$$(1) \quad [x \otimes t^n + \alpha c, y \otimes t^m + \beta c] := [x, y] \otimes t^{n+m} + \langle x, y \rangle m \delta_{m, -n} c,$$

for $x, y \in \mathfrak{g}$ and $\alpha, \beta \in \mathbb{C}$. This defines a Lie algebra structure on $\hat{\mathfrak{g}}$. Indeed, if $i: \mathbb{C} \rightarrow \hat{\mathfrak{g}}$ and $p: \hat{\mathfrak{g}} \rightarrow \mathcal{L}(\mathfrak{g})$ are the Lie algebra homomorphisms given by $\lambda \mapsto \lambda c$ and $x \otimes f + \alpha c \mapsto x \otimes f$ respectively, then $\hat{\mathfrak{g}}$ fits into the exact sequence

$$0 \longrightarrow \mathbb{C} \xrightarrow{i} \hat{\mathfrak{g}} \xrightarrow{p} \mathcal{L}(\mathfrak{g}) \longrightarrow 0.$$

Since $\hat{\mathfrak{g}}$ is simple, there exists only one non-degenerate invariant symmetric bilinear form on $\mathfrak{g}$ (up to scalar multiplication) given by the Killing form. Thus $\hat{\mathfrak{g}}$ is the unique central extension of $\mathcal{L}(\mathfrak{g})$.

Finally, denote by $\tilde{\mathfrak{g}}$ the Lie algebra obtained by adjoining to $\hat{\mathfrak{g}}$ a derivation d which has bracket (1) with the added relation $[c, d] = 0$ and

$$[d, x \otimes f] := x \otimes t \left(\frac{d}{dt} f \right).$$

Definition 2.2. The *affine Lie algebra of $\mathfrak{g}$* is defined as

$$\tilde{\mathfrak{g}} := \mathcal{L}(\mathfrak{g}) \oplus \mathbb{C}c \oplus \mathbb{C}d,$$

with Lie bracket

$$(2) \quad \begin{aligned} & [x \otimes t^n + \lambda_1 c + \mu_1 d, y \otimes t^m + \lambda_2 c + \mu_2 d] \\ &= ([x, y] \otimes t^{n+m} - \mu_2 n x \otimes t^n + \mu_1 m y \otimes t^m) + \langle x, y \rangle m \delta_{m, -n} c \end{aligned}$$

for $x, y \in \mathfrak{g}$ and $\lambda_1, \lambda_2, \mu_1, \mu_2 \in \mathbb{C}$.

Note that $\hat{\mathfrak{g}}$ and $\tilde{\mathfrak{g}}$ are both called affine Lie algebras in the literature, but we will be using this name for $\tilde{\mathfrak{g}}$.

Example 2.3. The loop algebra of $\mathfrak{sl}_2$ is

$$\mathcal{L}(\mathfrak{sl}_2) = \mathfrak{sl}_2 \otimes \mathbb{C}[t, t^{-1}]$$

which can be viewed as 2×2 traceless matrices with matrix elements in $\mathbb{C}[t, t^{-1}]$. For all $n \in \mathbb{Z}$, $\mathcal{L}(\widetilde{\mathfrak{sl}_2})$ has the basis

$$e \otimes t^n = \begin{pmatrix} 0 & t^n \\ 0 & 0 \end{pmatrix}, \quad h \otimes t^n = \begin{pmatrix} t^n & 0 \\ 0 & -t^n \end{pmatrix}, \quad f_n \otimes t^n = \begin{pmatrix} 0 & 0 \\ t^n & 0 \end{pmatrix}.$$

The affine lie algebra $\widetilde{\mathfrak{sl}_2} = \mathcal{L}(\mathfrak{sl}_2) \oplus \mathbb{C}c \oplus \mathbb{C}d$ is thus spanned by $\{x \otimes t^n + c + d \mid x \in \{e, h, f\}, n \in \mathbb{Z}\}$.

3. ROOT SPACE DECOMPOSITION

Now we may analyse the structure of affine Lie algebras just as we can with usual Lie theory. Let us start with defining Cartan subalgebras of $\hat{\mathfrak{g}}$.

Definition 3.1. The *Cartan subalgebra* $\tilde{\mathfrak{h}}$ associated to $\mathfrak{h}$ is the subalgebra defined by

$$\tilde{\mathfrak{h}} := \mathfrak{h} \oplus \mathbb{C}c \oplus \mathbb{C}d,$$

where $\mathfrak{h}$ is identified with $\mathfrak{h} \otimes 1$.

We can also extend the dual elements of $\mathfrak{h}$ to that of $\tilde{\mathfrak{h}}$. For each $\alpha \in \Delta \subset \mathfrak{h}^*$, let $\alpha(c) = \alpha(d) = 0$.

Definition 3.2. Define the *dual Cartan subalgebra* $\tilde{\mathfrak{h}}^*$ to be

$$\tilde{\mathfrak{h}}^* := \mathfrak{h}^* \oplus \mathbb{C}\gamma \oplus \mathbb{C}\delta,$$

where

$$\begin{aligned} \gamma(\mathfrak{h}) &:= 0, \gamma(c) := 1, \gamma(d) := 0 \\ \delta(\mathfrak{h}) &:= 0, \delta(c) := 0, \delta(d) := 1. \end{aligned}$$

Elements of $\tilde{\mathfrak{h}}^*$ are called *weights*.

This gives us a notion of affine roots and root spaces. For $\alpha \in \tilde{\mathfrak{h}}^*$, let

$$\tilde{\mathfrak{g}}_\alpha := \{x \in \tilde{\mathfrak{g}} \mid [\tilde{h}, x] = \alpha(\tilde{h})x \text{ for all } \tilde{h} \in \tilde{\mathfrak{h}}\}$$

be the root space associated to α .

Proposition 3.3. *There is a decomposition*

$$\tilde{\mathfrak{g}} = \tilde{\mathfrak{h}} \oplus \bigoplus_{\beta \in \tilde{\Phi}} \tilde{\mathfrak{g}}_\beta,$$

where $\tilde{\Phi} := \{\alpha + m\delta \mid \alpha \in \Phi, m \in \mathbb{Z}\} \cup \{m\delta \mid m \in \mathbb{Z} \setminus \{0\}\}$.

Proof. For each $\alpha \in \Phi$, fix a nonzero $x_\alpha \in \mathfrak{g}_\alpha$. Then for all $\alpha \in \Phi$, $m \in \mathbb{Z}$, and any $\tilde{h} = h \otimes 1 + \lambda c + \mu d \in \tilde{\mathfrak{h}}$, we get that

$$\begin{aligned} [\tilde{h}, x_\alpha \otimes t^m] &= [h \otimes 1 + \lambda c + \mu d, x_\alpha \otimes t^m] \\ &= [h, x_\alpha] \otimes t^m + \mu m x_\alpha \otimes t^m + (h, x_\alpha) m \delta_{m,0} c \\ &= \alpha(h) x_\alpha \otimes t^m + m \delta(\tilde{h}) x_\alpha \otimes t^m \\ &= (\alpha + m\delta)(\tilde{h}) \cdot (x_\alpha \otimes t^m), \end{aligned}$$

since $(h, x_\alpha) m \delta_{m,0} c = 0$ for all $m \in \mathbb{Z}$. This tells us that $\mathfrak{g}_\alpha \otimes t^m \subseteq \tilde{\mathfrak{g}}_{\alpha+m\delta}$ for $\alpha \in \Phi$. The other inclusion holds similarly.

Next for each $h' \in \mathfrak{h}$ and $m \in \mathbb{Z} \setminus \{0\}$, it follows that

$$\begin{aligned} [\tilde{h}, h' \otimes t^m] &= [h \otimes 1 + \lambda c + \mu d, h' \otimes t^m] \\ &= [h, h'] \otimes t^m + \mu m h \otimes t^m + (h, h') m \delta_{m,0} c \\ &= (m\delta(\tilde{h})) \cdot (h' \otimes t^m), \end{aligned}$$

where $[h, h'] = 0$ as $\mathfrak{h}$ is abelian. This gives $h \otimes t^m \subseteq \tilde{\mathfrak{g}}_{m\delta}$, and again, the reverse inclusion holds similarly. $\square$

The standard choice of positive roots is

$$\tilde{\Phi}^+ := \{\alpha + m\delta \mid \alpha \in \Phi, m \in \mathbb{Z}_{>0}\} \cup \{m\delta \mid m \in \mathbb{Z}_{>0}\} \cup \Phi^+,$$

from which it is easy to see that the simple roots must be

$$\tilde{\Delta} := \{\delta - \theta\} \cup \Delta,$$

where θ is the highest root of $\mathfrak{g}$ with respect to the partial ordering on $\mathfrak{h}^*$. Similarly define $\tilde{\Phi}^- := -\tilde{\Phi}^+$. This decomposition of roots allows us to define the subalgebras

$$\tilde{\mathfrak{n}}^\pm := \bigoplus_{\alpha \in \tilde{\Phi}^\pm} \tilde{\mathfrak{g}}_\alpha,$$

giving rise to a triangular decomposition

$$\tilde{\mathfrak{g}} = \tilde{\mathfrak{n}}^- \oplus \tilde{\mathfrak{h}} \oplus \tilde{\mathfrak{n}}^+.$$

Remark. We call the roots $\tilde{\Delta}^{\text{re}} := \{\alpha + m\delta \mid m \in \mathbb{Z}, \alpha \in \Delta\}$ *real* and the roots $\tilde{\Delta}^{\text{im}} := \{m\delta \mid m \in \mathbb{Z} \setminus \{0\}\}$ *imaginary*. Note that this naming is not related to the complex numbers, but the fact that the lengths of imaginary roots with respect to the inner product is non-positive, whereas the lengths of the real roots are positive. See [Kac90, Chapter 5] for more details.

Example 3.4. Figure 2 shows the roots for $\widetilde{\mathfrak{sl}}_2$. The simple roots are $\{\delta - \alpha, \alpha\}$ indicated by the arrows. The positive roots are the roots on the right of and including α , δ and $\delta - \alpha$, and the negative roots are the remaining non-zero roots. The n -th column contains the roots for $\mathfrak{g} \otimes t^n$. The roots denoted by the open circles are the imaginary roots while the filled in circles denote the real roots.

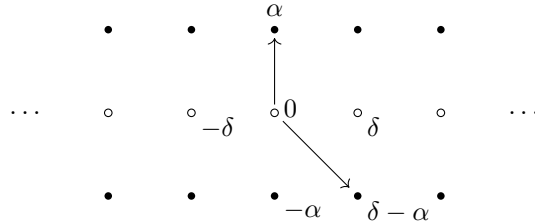


FIGURE 2. Roots of $\widetilde{\mathfrak{sl}}_2$

4. HIGHEST WEIGHT REPRESENTATIONS OF $\tilde{\mathfrak{g}}$

Just as in the representation theory of standard Lie algebras, highest weight representations of affine Lie algebras may be defined. We first introduce an important class of $\tilde{\mathfrak{g}}$ -modules obtained from $\mathfrak{g}$ -modules via induction¹.

¹We take for granted the theory of universal enveloping algebras for affine Lie algebras (and more generally for Kac-Moody algebras), so that induction is well-defined (see [Kac90, 9.1]).

4.1. Verma modules.

Definition 4.1. Given a weight $\Lambda \in \tilde{\mathfrak{h}}^*$, the *Verma module associated to Λ* is

$$M_\Lambda := \text{Ind}_{\tilde{\mathfrak{h}} \oplus \tilde{\mathfrak{n}}^+}^{\tilde{\mathfrak{g}}} \mathbb{C}_\Lambda = U(\tilde{\mathfrak{g}}) \otimes_{U(\tilde{\mathfrak{h}} \oplus \tilde{\mathfrak{n}}^+)} \mathbb{C}_\Lambda,$$

where $\mathbb{C}_\Lambda$ is a one-dimensional representation of $\tilde{\mathfrak{h}} \oplus \tilde{\mathfrak{n}}^+$ on which $\tilde{\mathfrak{n}}^+$ acts by zero and $h \in \tilde{\mathfrak{h}}$ acts by the scalar $\Lambda(h)$.

We say that V is a *highest weight representation of weight Λ* of $\tilde{\mathfrak{g}}$ if it is generated by a vector v_Λ with weight Λ and $\tilde{\mathfrak{n}}^+ \cdot v_\Lambda = 0$.

For the rest of the paper, we will restrict² to weights $\Lambda = \lambda + k\gamma - \Delta\delta$ such that

$$\Delta = \Delta(\lambda) = \frac{\langle \lambda, \lambda + 2\rho \rangle}{2(k + h^\vee)},$$

where $h^\vee$ is the dual Coxeter number (see Appendix A).

Verma modules with such a weight Λ will be henceforth denoted $M_{\lambda,k}$.

Lemma 4.2. The Verma module $M_{\lambda,k}$ decomposes into $\mathfrak{g}$ -subrepresentations as

$$M_{\lambda,k} = \bigoplus_{n \geq 0} M_{\lambda,k}[-n],$$

where $M_{\lambda,k}[-n]$ is the eigenspace of the action of d with eigenvalue $-n - \Delta(\lambda)$.

We give an example illustrating this lemma.

Example 4.3. Let us consider the $\widetilde{\mathfrak{sl}}_2$ -module $M_{\lambda,k}$ for $\lambda = \alpha/2$ and $k = 1$. Then $\rho = \alpha/2$ and $h^\vee = 2$ (since $\mathfrak{sl}_2$ has type A_1), so

$$\Delta(\alpha) = \frac{\langle \frac{\alpha}{2}, \frac{3\alpha}{2} \rangle}{2(1+2)} = \frac{\frac{3}{4} \cdot \langle \alpha, \alpha \rangle}{6} = \frac{\frac{3}{4} \cdot 2}{6} = \frac{1}{4}$$

Then $\Lambda = \frac{1}{2}\alpha + \gamma - \frac{1}{4}\delta$. Figure 3 gives a visualisation of the weight space of $M_{\alpha,1}$.

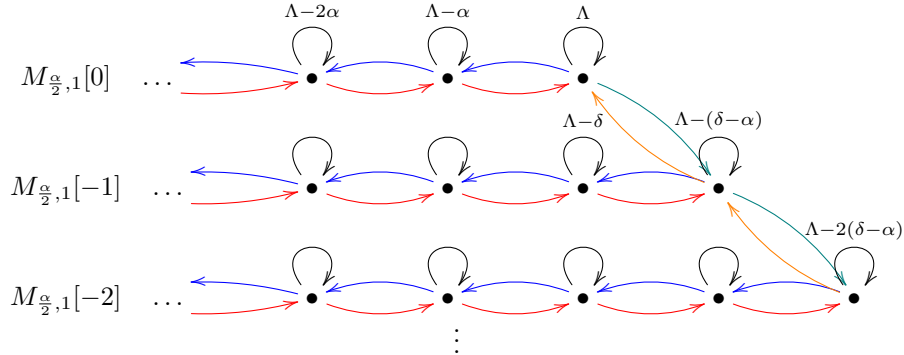


FIGURE 3. Weight space of $M_{\frac{\alpha}{2},1}$

²This choice for the coefficient of δ is motivated from the construction of the Segal-Sugawara operators, which define an action of the Virasoro algebra on certain highest-weight representations of $\hat{\mathfrak{g}}$ (see [EFK98, 2.9]). These operators are essential in defining the Knizhnik-Zamolodchikov equations which play an important role in WZNW conformal field theory.

Let v_Λ be the highest weight vector of $M_{\frac{\alpha}{2},1}$, which generates the top-right weight space. We claim that the action of $f \otimes t^0$ on v_Λ generates $M_{\frac{\alpha}{2},1}[0]$. The vector $(f \otimes t^0) \cdot v_\Lambda$ has weight $\Lambda - \alpha$ since

$$\begin{aligned} h \cdot ((f \otimes t^0) \cdot v_\Lambda) &= [h \otimes t^0, f \otimes t^0] \cdot v_\Lambda + (f \otimes t^0) \cdot (h \cdot v_\Lambda) \\ &= ([h, f] \otimes t^0) \cdot v_\Lambda + \Lambda(h)(f \otimes t^0) \cdot v_\Lambda \\ &= -2(f \otimes t^0) \cdot v_\Lambda + \frac{\alpha}{2}(h)(f \otimes t^0) \cdot v_\Lambda \\ &= -(f \otimes t^0) \cdot v_\Lambda \\ &= (\Lambda - \alpha)(h)(f \otimes t^0) \cdot v_\Lambda, \end{aligned}$$

and this can be extended to any $\tilde{h} \in \tilde{\mathfrak{h}}$ by the commutation relations. The action of d keeps $(f \otimes t^0) \cdot v_\Lambda$ in $M_{\alpha,1}[0]$ since

$$d \cdot ((f \otimes t^0) \cdot v_\Lambda) = [d, f \otimes t^0] \cdot v_\Lambda + (f \otimes t^0) \cdot (d \cdot v_\Lambda) = -\Delta\left(\frac{\alpha}{2}\right)(f \otimes t^0) \cdot v_\Lambda.$$

It follows from an induction argument that $(f \otimes t^0)^k \cdot v_\Lambda$ has weight $\Lambda - k\alpha$. Here, $M_{\frac{\alpha}{2},1}[0]$ is isomorphic to a highest weight module $M_{\frac{\alpha}{2}}$.

Now consider the action of $e \otimes t^{-1}$ on v_Λ . For $\tilde{h} = h + \mu_1 c + \mu_2 d$, we have that

$$\begin{aligned} \tilde{h} \cdot ((e \otimes t^{-1}) \cdot v_\Lambda) &= [h + \mu_1 c + \mu_2 d, e \otimes t^{-1}] \cdot v_\Lambda + (e \otimes t^{-1}) \cdot (\tilde{h} \cdot v_\Lambda) \\ &= [h \otimes t^0, e \otimes t^{-1}] \cdot v_\Lambda - \mu_2(e \otimes t^{-1}) \cdot v_\Lambda + \Lambda(\tilde{h})(e \otimes t^{-1}) \cdot v_\Lambda \\ &= -2(e \otimes t^{-1}) \cdot v_\Lambda - \mu_2(e \otimes t^{-1}) \cdot v_\Lambda + \Lambda(\tilde{h})(e \otimes t^{-1}) \cdot v_\Lambda \\ &= (\Lambda - \delta + \alpha)(\tilde{h})(e \otimes t^{-1}) \cdot v_\Lambda, \end{aligned}$$

so $(e \otimes t^{-1}) \cdot v_\Lambda$ has weight $\Lambda - \delta + \alpha$. By induction, $(e \otimes t^{-1})^k \cdot v_\Lambda$ has weight $\Lambda - k(\delta - \alpha)$. Now let us see how we can take v_Λ down d -graded components. Take $(e \otimes t^{-1}) \cdot v_\Lambda$. We get

$$\begin{aligned} d \cdot ((e \otimes t^{-1}) \cdot v_\Lambda) &= [d, e \otimes t^{-1}] \cdot v_\Lambda + (e \otimes t^{-1}) \cdot (d \cdot v_\Lambda) \\ &= -(e \otimes t^{-1}) \cdot v_\Lambda - \Delta\left(\frac{\alpha}{2}\right)(e \otimes t^{-1}) \cdot v_\Lambda \\ &= \left(-1 - \Delta\left(\frac{\alpha}{2}\right)\right)(e \otimes t^{-1}) \cdot v_\Lambda \end{aligned}$$

Again, one can see via induction that for $n > 0$, $e \otimes t^{-n} \cdot v_\Lambda \in M_{\frac{\alpha}{2},1}[-n]$. Analogously, $f \otimes t^1$ raises the d -grading and subtracts α from the weight.

In Figure 3, the action $f \otimes t^0$ corresponds to the blue arrows and $e \otimes t^0$ the red arrows since each $M_{\frac{\alpha}{2},1}[-n]$ is a $\mathfrak{g}$ -representation. The green arrows corresponds to the action $e \otimes t^{-1}$ which shifts the d -degree down and one to the right, while the orange arrow corresponds to the action $f \otimes t$ which raises the d -degree and shifts one to the left.

Compare this diagram with Figure 2, which shows the roots of $\widetilde{\mathfrak{sl}}_2$ being spanned by α and $\delta - \alpha$. Here in the weight spaces of $M_{\frac{\alpha}{2},1}$, the action by the blue arrows decrease the weight by α and the action of the green arrows decrease the weight by $\delta - \alpha$. We also see that each level is isomorphic to a Verma module of $\mathfrak{sl}_2$ as in Figure 1.

The following theorem gives a criterion for when M_Λ is irreducible.

Theorem 4.4 ([KK79, Proposition 3.1]). *The Verma module M_Λ is irreducible if and only if for every $\alpha \in \tilde{\Phi}^+$ and all $n \in \mathbb{Z}_{>0}$,*

$$\langle \Lambda + \hat{\rho}, \alpha \rangle \neq \frac{n}{2} \langle \alpha, \alpha \rangle,$$

where $\hat{\rho} = \rho + h^\vee \gamma$.

Weights Λ for which M_Λ is irreducible are called *generic*.

Example 4.5. For $M_{\frac{\alpha}{2}, 1}$, we have that $\hat{\rho} = \frac{\alpha}{2} + 2\gamma$. Taking the positive root $\alpha \in \tilde{\Phi}^+$, we get that

$$\langle \Lambda + \hat{\rho}, \alpha \rangle = \left\langle \alpha + 3\gamma - \frac{1}{4}\delta, \alpha \right\rangle = \langle \alpha, \alpha \rangle,$$

as $\langle \gamma, \alpha \rangle = \langle \delta, \alpha \rangle = 0$ are trivial pairs. This means that for the equation in Theorem 4.4 holds for $n = 2$. Therefore, $M_{\frac{\alpha}{2}, 1}$ is reducible.

4.2. Weyl modules. Since there is a unique maximal proper submodule of M_Λ just like in the classical case (see [Hum21, 1.3]), there is a unique irreducible quotient L_Λ of M_Λ , which we denote by $L_{\lambda, k}$ when $\Delta = \Delta(\lambda)$. By Theorem 4.4, $L_\Lambda = M_\Lambda$ when Λ is generic.

Another important example of highest weight representations are those induced from $L_{\lambda, k}$. Let $\tilde{\mathfrak{g}}^+ := \mathfrak{g} \otimes \mathbb{C}[t] \oplus \mathbb{C}c \oplus \mathbb{C}d$. If V is a module over $\mathfrak{g}$, then it is automatically a module over $\tilde{\mathfrak{g}}$ where $\mathfrak{g} \otimes t^0$ is identified with $\mathfrak{g}$, $\mathfrak{g} \otimes t\mathbb{C}[t]$ acts by zero, and c, d by constants $k, -\Delta \in \mathbb{C}$ respectively.

Definition 4.6. Let λ be a dominant integral weight of $\mathfrak{g}$ and let $k \in \mathbb{C}$. The *Weyl module* $V_{\lambda, k}$ is defined as

$$V_{\lambda, k} := \text{Ind}_{\tilde{\mathfrak{g}}^+}^{\tilde{\mathfrak{g}}} L_{\lambda, k},$$

where d acts by multiplication by $-\Delta(\lambda)$ in L_λ .

Example 4.7. For $\lambda = 0$ and $k = 1$, $\Lambda = \gamma$ and $M_{0, 1}$ has a quotient $V_{0, 1}$. Here, $\Delta(0) = 0$. Using a similar methods in Example 4.3, we find that the simple quotient is an infinite-dimensional highest weight module with the following weight space:

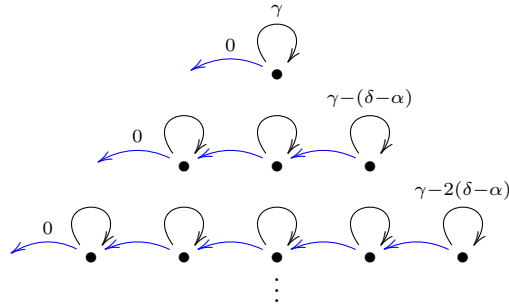


FIGURE 4. Weight space of $V_{0,1}$

The blue arrows correspond to action by $f \otimes t^0$, which decreases the weight by α . Each d -graded component is finite-dimensional, generated by the vector in the right-most weight space under the action of $f \otimes t^0$. Although not depicted in the figure, the action of $e \otimes t^0$ increases the weight in the same level, and $- \otimes t^n$ decreases the degree by n .

Lastly, we state a theorem that characterises irreducible Weyl modules.

Theorem 4.8 ([KK79, Theorem 2]). *Let $\lambda \in \mathfrak{h}^*$ and $k \in \mathbb{C}$. Then $V_{\lambda,k}$ is irreducible if for all $\alpha \in \Phi^+$, $k \notin \mathbb{Q}\langle\alpha^\vee, \lambda\rangle + \mathbb{Q}$.*

An immediate corollary of this theorem is that $V_{\lambda,k}$ is irreducible if k is irrational. For such k , we thus get $V_{\lambda,k} = L_{\lambda,k}$.

APPENDIX A.

Definition A.1. Let $\theta = \sum_i \chi_i \alpha_i$ be the maximal root of $\mathfrak{g}$ where the α_i are simple roots. The *dual Coxeter number* $h^\vee$ is defined as

$$h^\vee := 1 + \sum_i \chi_i^\vee = 1 + \langle \rho, \theta \rangle,$$

where $\chi_i^\vee = \chi_i \frac{\langle \alpha_i, \alpha_i \rangle}{2}$.

Below is a list of dual Coxeter numbers in classical type.

A_n	B_n	C_n	D_n	E_6	E_7	E_8	F_4	G_2
$n+1$	$2n-1$	$n+1$	$2n-2$	12	18	30	9	4

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